

KINOVA GEN3 PHYSICS REPLICATION IN MATLAB/SIMULINK USING SIMSCAPE MULTIBODY

INTRODUCTION

The primary objective of this task was to implement and validate a physics-based simulation of the Kinova Gen3 robotic manipulator using MATLAB and Simulink. The focus was to transition from basic kinematic or motion-driven simulation to a more realistic representation of the system by incorporating physical forces, specifically torque, as the driving input. In robotic systems, accurate physics modelling is essential for understanding real-world behaviour, validating control strategies, and ensuring system reliability. This task forms a foundational step toward achieving a fully integrated simulation environment that can later be extended to platforms such as NVIDIA Isaac Sim.

TOOLS AND ENVIRONMENT

The implementation was carried out using MATLAB in a cloud-based environment, along with Simulink and Simscape Multibody. The Robotics System Toolbox was used to access the Kinova Gen3 model and its predefined kinematic structure. Simscape Multibody served as the core physics engine, enabling simulation of rigid body dynamics, joint constraints, and force-based interactions. The combination of these tools allowed seamless integration between control signals and physical simulation, making it suitable for modelling complex robotic systems.

MODEL GENERATION AND SETUP

The Kinova Gen3 robotic model was imported using MATLAB's built-in function, which provides a complete rigid body representation of the manipulator. The model consists of seven revolute joints and one fixed end-effector, forming a complete kinematic chain. The structure was verified to ensure that all links, joints, and connections were correctly defined. Using the Simulink import functionality, the robot was converted into a block-based Simscape Multibody model. This generated a visual representation where each joint is modelled as an actuator block, and each link represents a rigid body with associated physical properties such as mass, inertia, and centre of mass. This setup provided a ready-to-use physical system for further simulation and testing.

INITIAL MOTION-BASED VALIDATION

Before implementing force-based control, the model was first validated using a motion-driven approach. A Step input signal was applied to a selected joint (Actuator4), and the resulting position output was observed using Scope blocks. This step ensured that the signal flow between Simulink and Simscape was correctly established and that the joint could respond to input commands. The results showed a clean and predictable step response, confirming that the model was functioning correctly at a basic level. However,

this approach directly imposed motion on the joint and did not reflect realistic physical behaviour, as it bypassed the dynamics of force and acceleration.

IMPLEMENTATION OF TORQUE-BASED PHYSICS

To achieve a realistic simulation, the system was modified to use torque as the input rather than direct motion. The actuator configuration was updated so that torque was provided as an external input, while motion was computed automatically by the system. This change allowed the joint to behave according to physical laws rather than following a predefined motion trajectory. A Constant input signal was used to represent applied torque, and this signal was passed through a Simulink-PS Converter to convert it into a physical quantity measured in Newton-metres. The torque was then applied to the actuator's torque input port, enabling force-driven simulation.

OUTPUT MEASUREMENT AND ANALYSIS

To analyse the system's behaviour, both position and velocity outputs were measured from the actuator. The position output represents the angular displacement of the joint, while the velocity output represents the rate of change of that displacement. These signals were converted back into Simulink signals using PS-Simulink Converters and visualised using Scope blocks. The simulation results showed a dynamic response when torque was applied. Specifically, the velocity exhibited an initial spike, followed by oscillations and gradual stabilisation. The position output reflected corresponding changes over time. This behaviour is characteristic of real physical systems, where applied torque leads to acceleration, and damping effects gradually stabilise the motion.

PHYSICS ENGINE AND SIMULATION BEHAVIOUR

The simulation was powered by Simscape Multibody, which acts as the physics engine. It models rigid body dynamics, joint interactions, and force propagation across the system. Unlike purely signal-based simulation, Simscape uses physical networks where variables represent real-world quantities such as force, torque, and velocity. This allows the system to compute motion based on underlying physical equations. The observed transient behaviour in the velocity graph, including oscillations and damping, confirms that the system is responding according to physical principles rather than artificially imposed motion. This validates the correctness of the physics implementation.

CHALLENGES AND RESOLUTIONS

Several challenges were encountered during the implementation process. Initially, the system produced flat outputs due to incorrect actuator configuration, which was resolved by properly setting torque as the input and motion as automatically computed. Connection issues, such as invalid or red signal lines, were addressed by ensuring correct unit assignment, particularly for torque in Newton-metres. Solver-related errors, especially those related to input derivatives, were resolved by configuring the input handling settings in the Simulink-PS Converter. Additionally, instability in the system response, such as excessive oscillations, was mitigated by reducing the torque magnitude and adjusting damping parameters within the actuator settings. These

challenges provided valuable insights into the interaction between control signals and physical simulation.

REFLECTION

This task significantly enhanced understanding of the difference between kinematic and dynamic simulation. While motion-based inputs provide a simplified way to test system connectivity, they do not accurately represent real-world behaviour. The transition to torque-based actuation highlighted the importance of force-driven simulation in robotics. It also demonstrated how system parameters such as damping, solver configuration, and input magnitude influence the stability and realism of the simulation. The process of debugging and resolving simulation errors contributed to the development of problem-solving skills and a deeper understanding of simulation environments. Overall, this task provided a practical foundation for implementing realistic robotic simulations.

SUMMARY

The Kinova Gen3 robot was successfully imported, configured, and simulated in MATLAB using Simulink and Simscape Multibody. Both motion-based and torque-based simulations were implemented and analysed. The final system demonstrates realistic physics-driven behaviour, with torque acting as the input and position and velocity as the outputs. The results confirm that the simulation accurately reflects dynamic system behaviour, making it suitable for further development and integration with advanced robotic simulation frameworks.

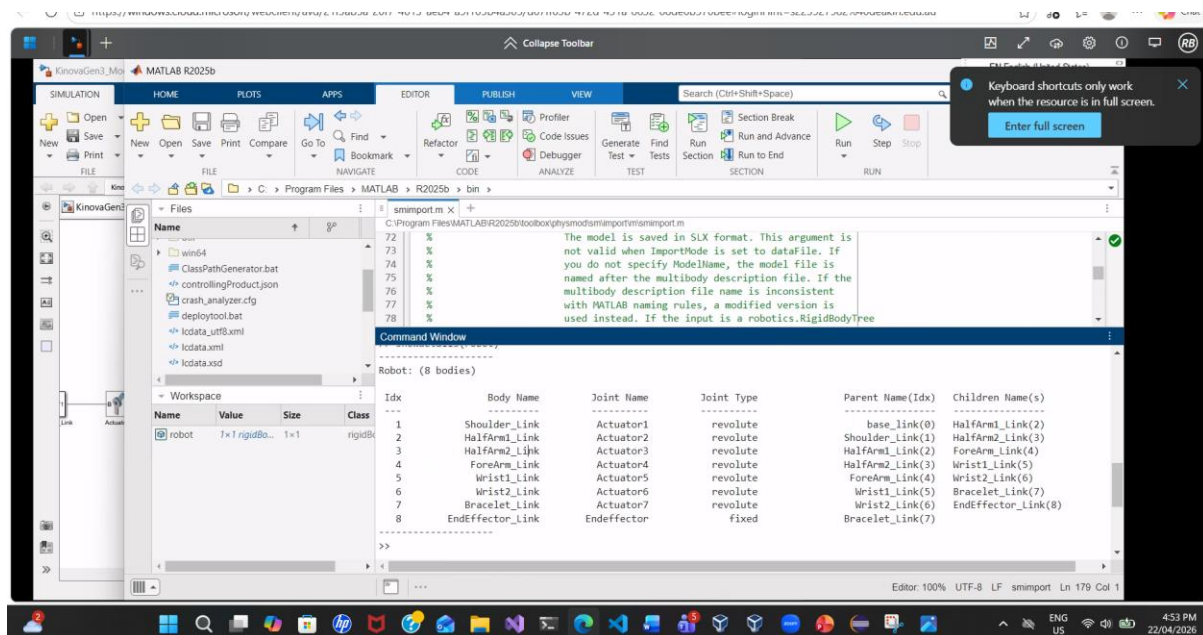
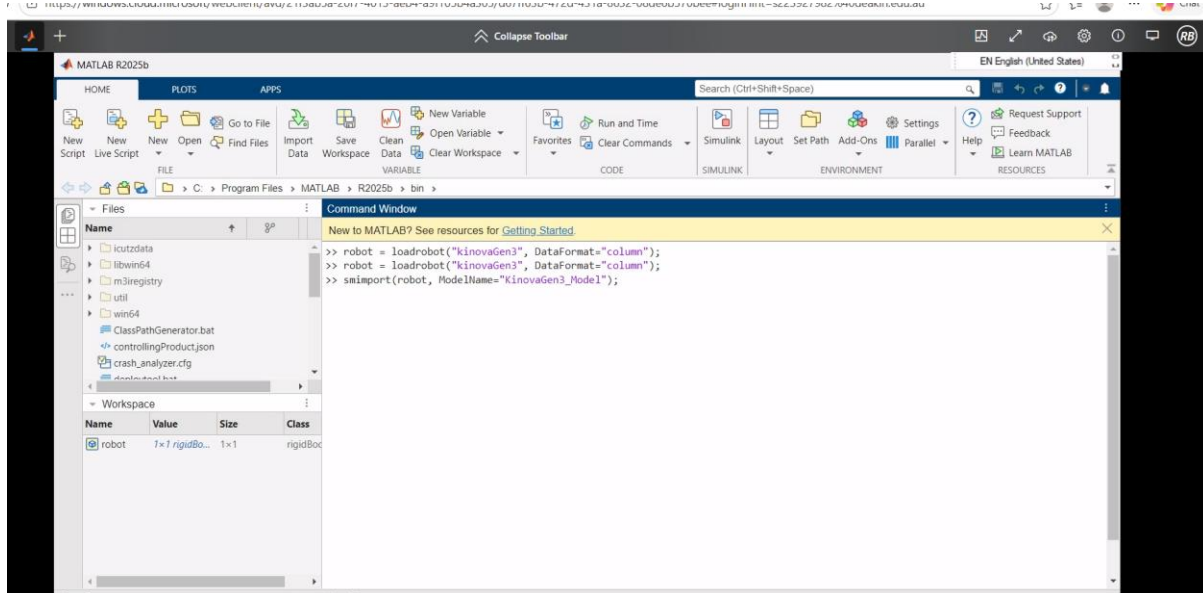
FUTURE WORK

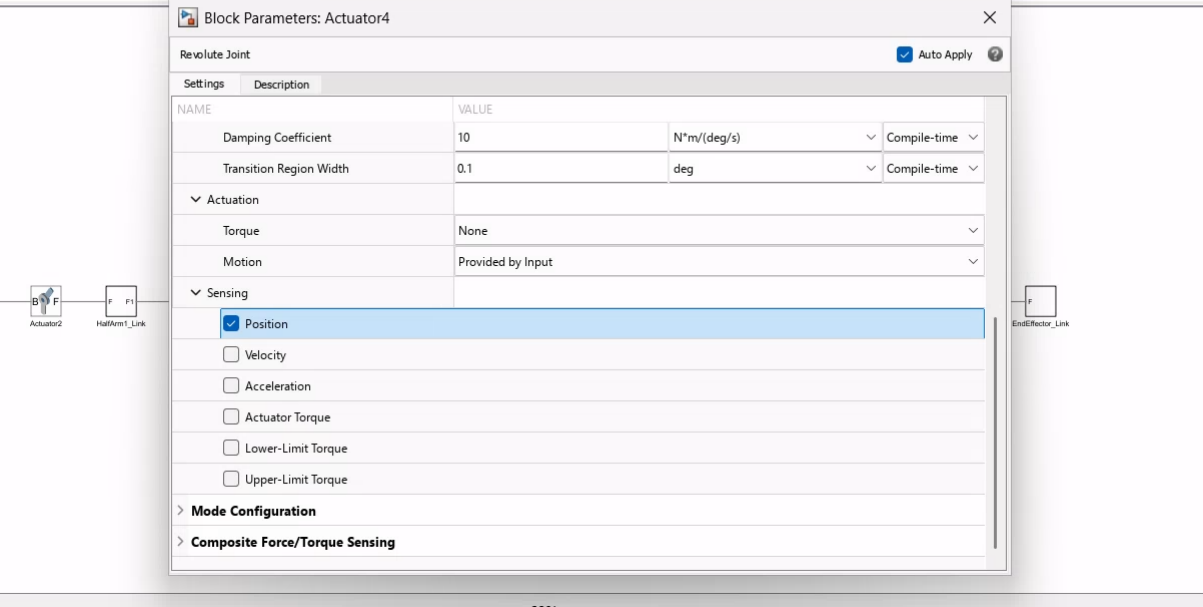
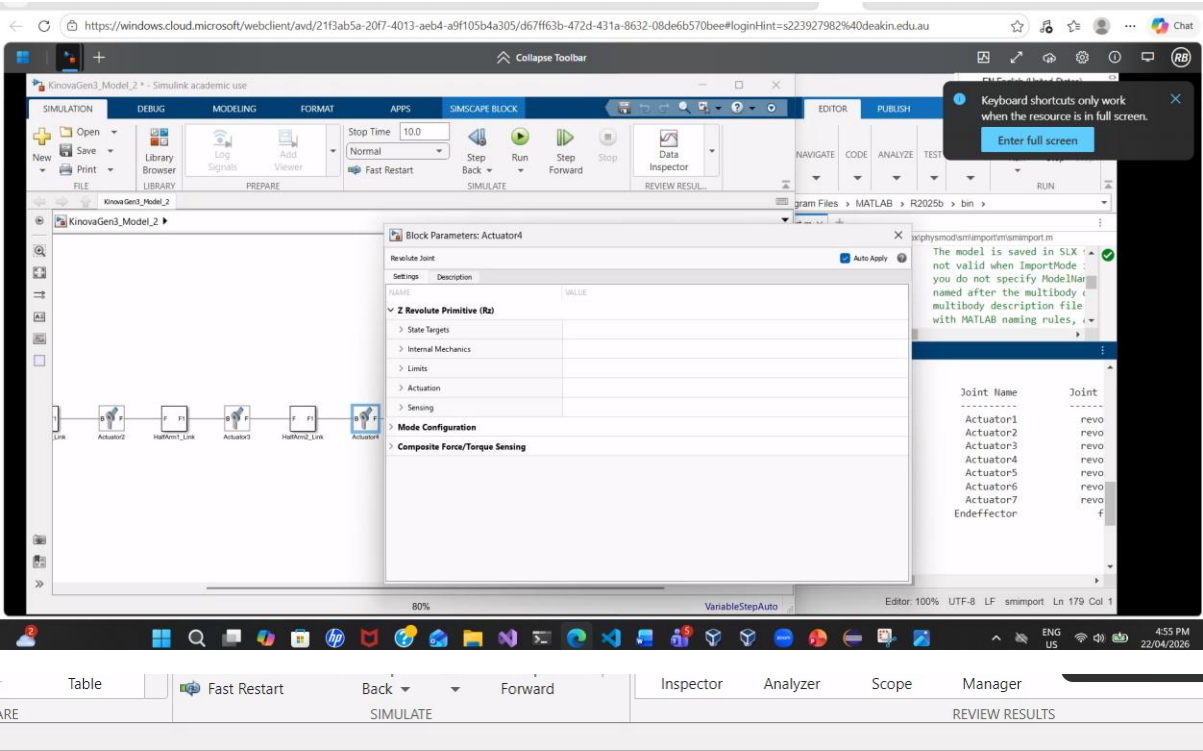
Future improvements include extending torque-based simulation to multiple joints, refining system stability through parameter tuning, and implementing control strategies such as PID or trajectory tracking. Integration with external simulation environments like Isaac Sim will also be explored to create a complete closed-loop robotic system. Additionally, incorporating perception and feedback mechanisms will further enhance the realism and applicability of the simulation.

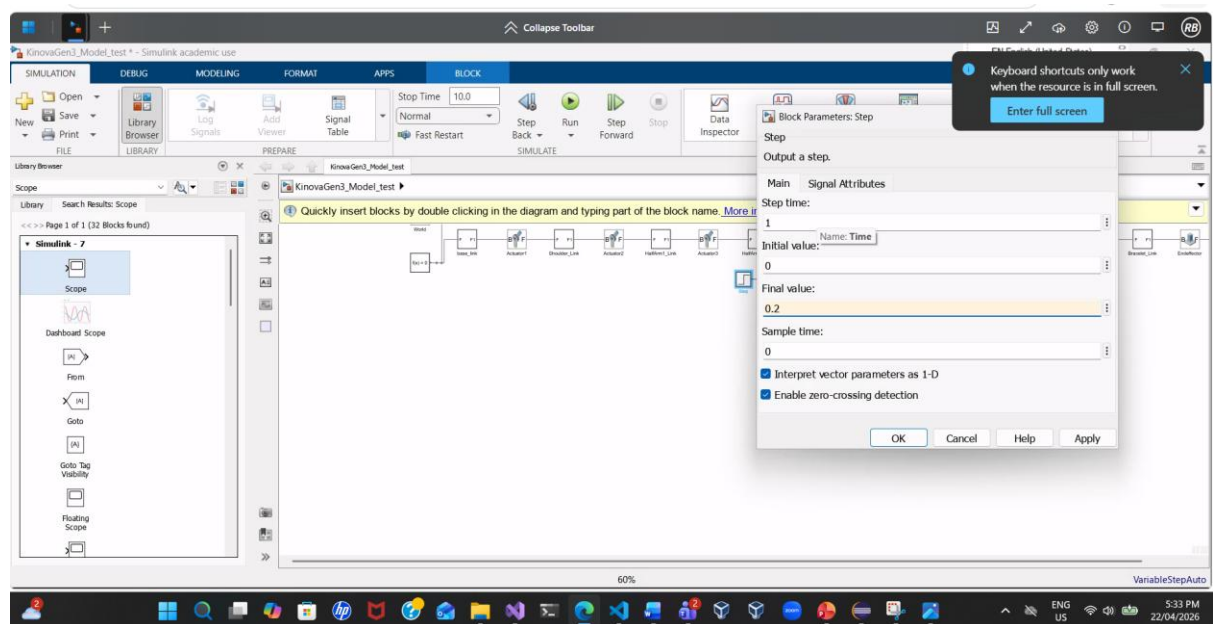
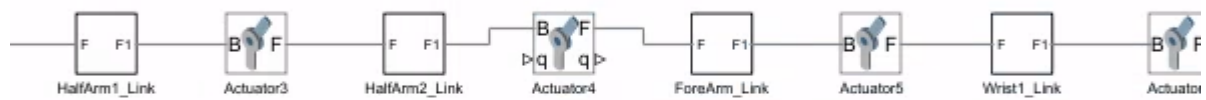
CONCLUSION

The implementation of Kinova Gen3 physics replication in MATLAB/Simulink has been successfully completed. The system now operates based on force-driven dynamics, with torque inputs producing realistic joint motion and measurable outputs. This marks a significant step toward developing a fully functional robotic simulation environment and provides a strong foundation for future work in control, perception, and system integration.

SCREENSHOTS



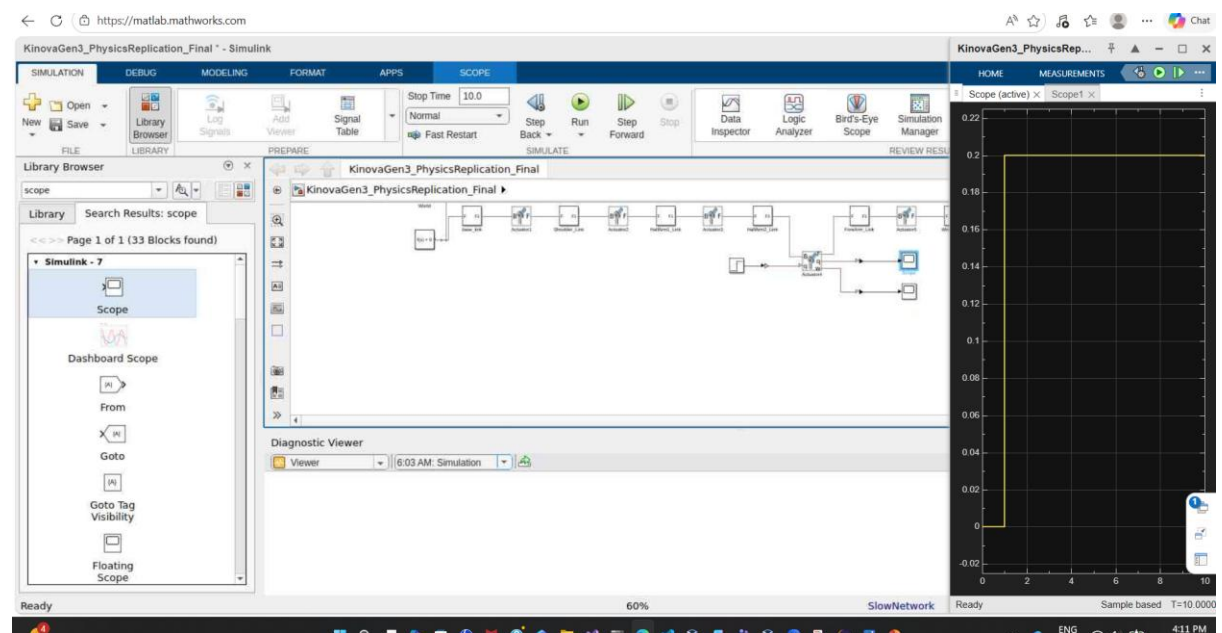
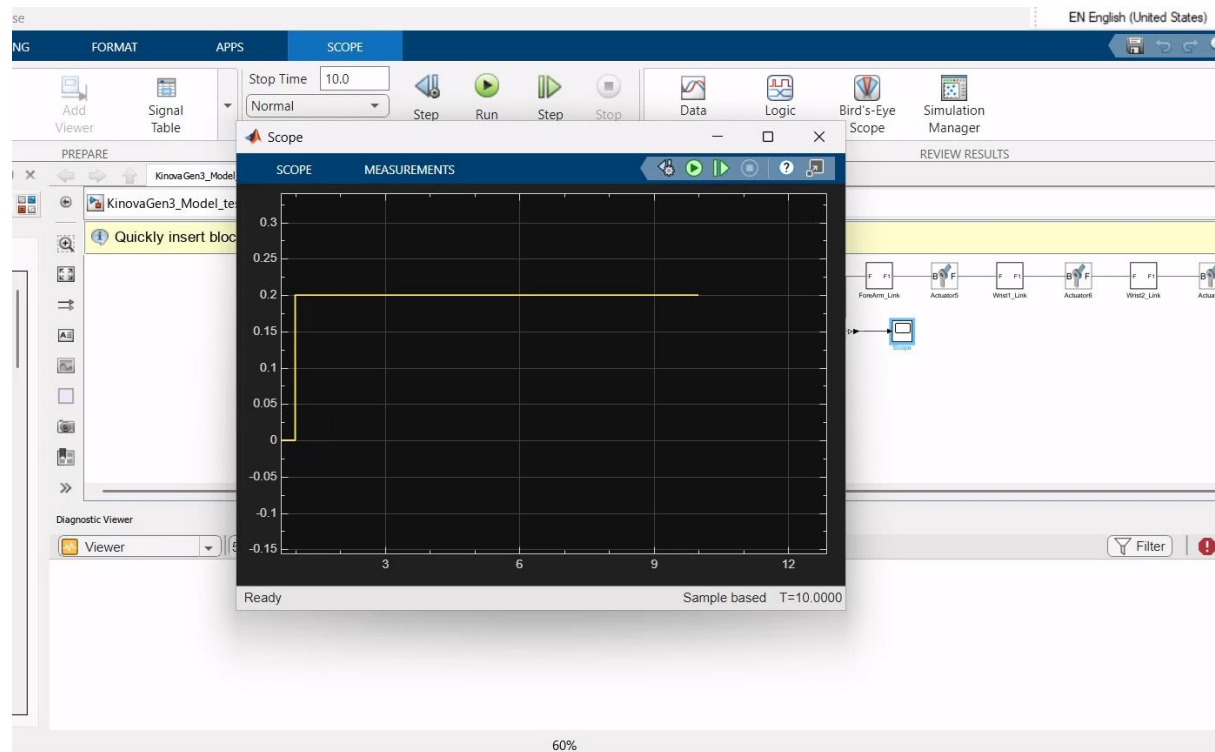




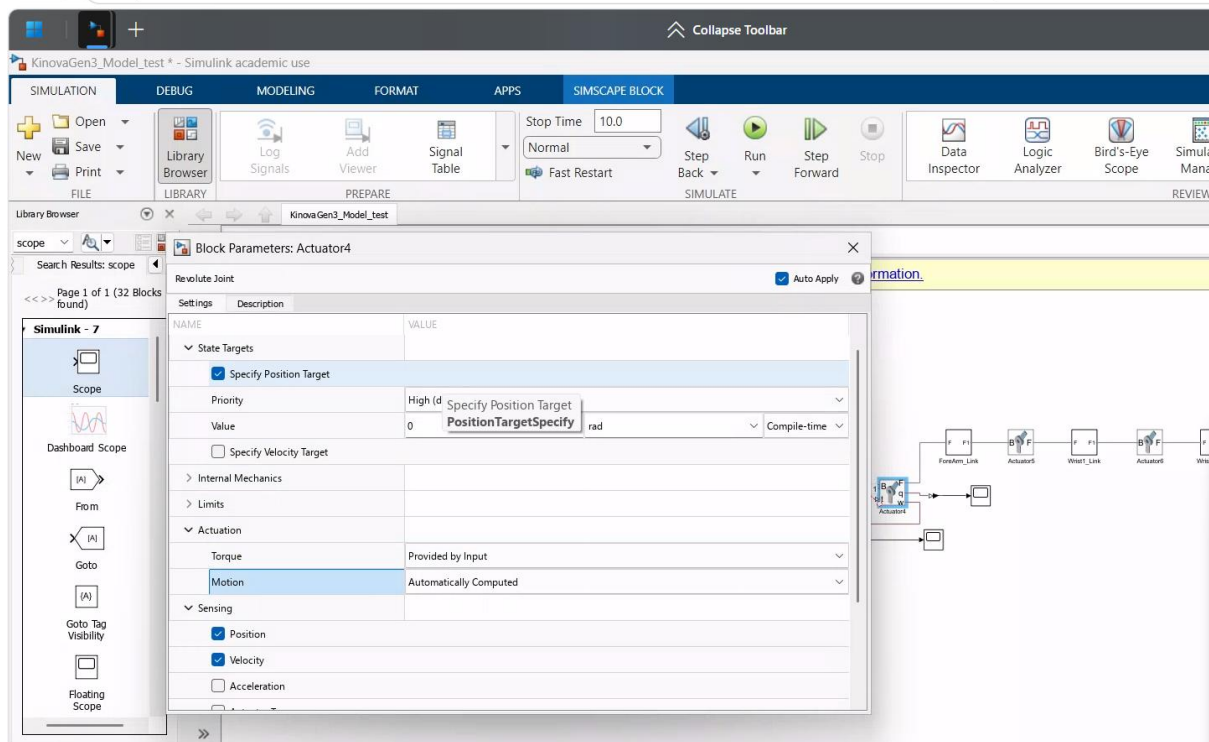
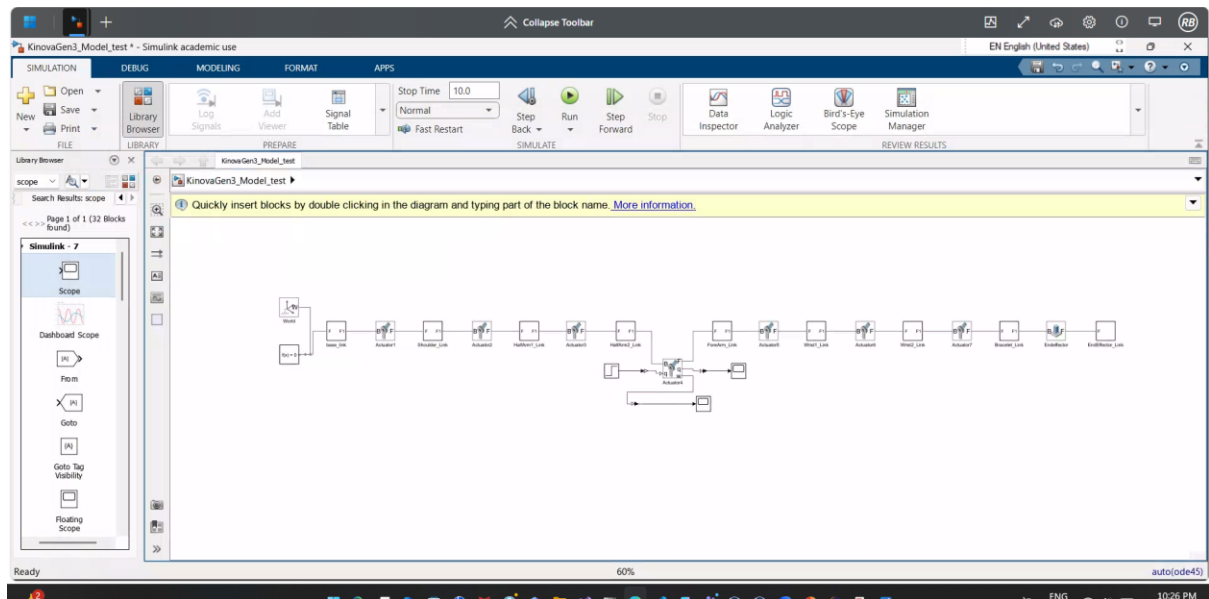
The image displays two screenshots from the MATLAB/Simulink environment. The top screenshot shows the 'Block Parameters: Simulink-PS Converter' dialog box, which is used to configure the parameters of the Simulink-PS Converter block. The dialog has two tabs: 'Settings' and 'Description'. The 'Settings' tab is active, showing a table of parameters and their values.

NAME	VALUE
Units	
Input signal unit	1
☐ Apply affine conversion	A
Input Handling	
Filtering and derivatives	m^3/s
Provided signals	Pa
	N
	m
	m/s
	N*m
	rad/s
	rad
	K

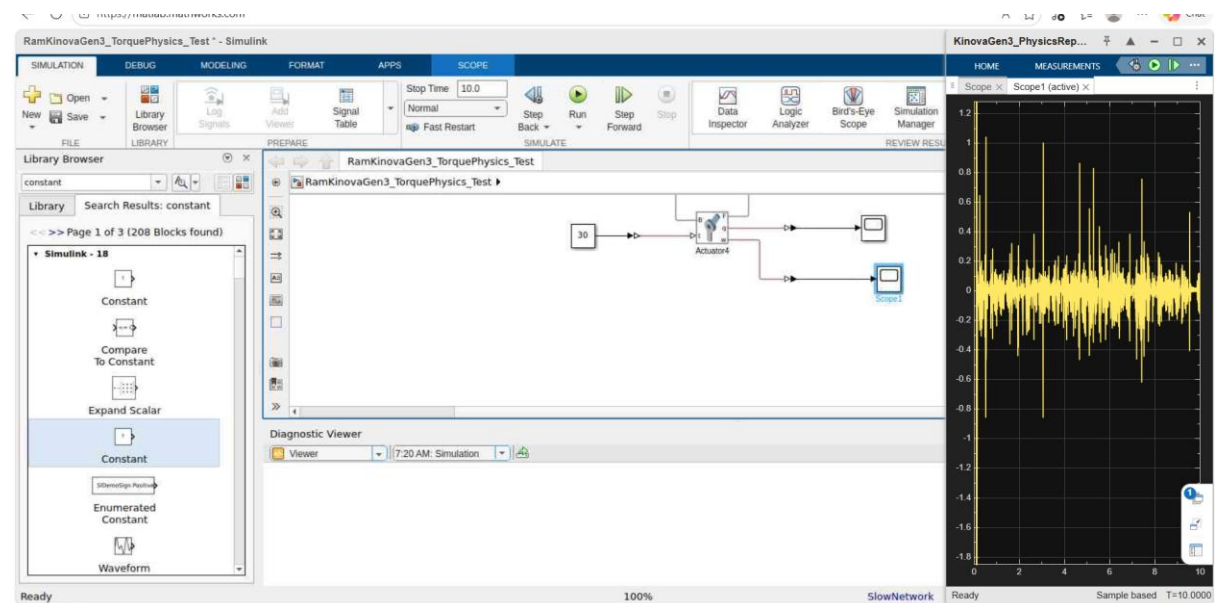
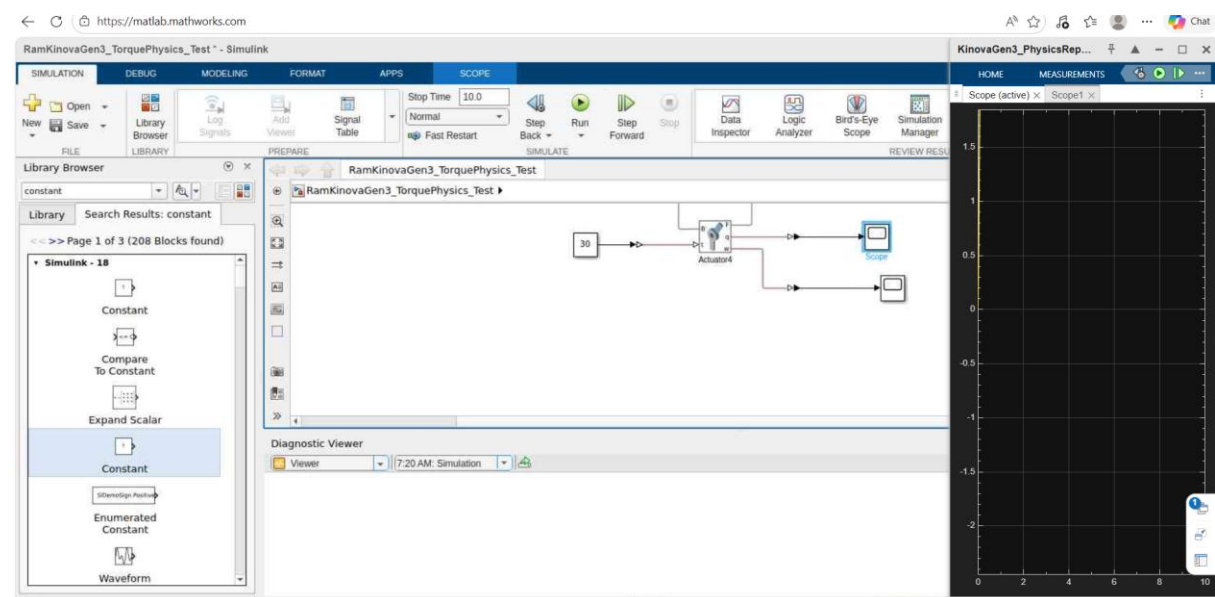
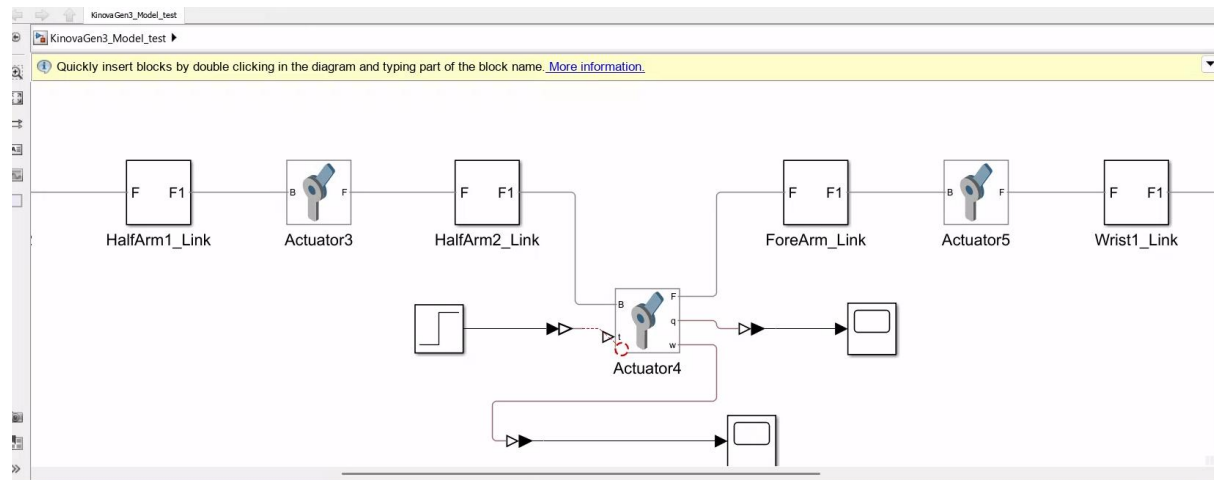
The bottom screenshot shows the main Simulink model window for 'KinovaGen3_Model_test'. The window includes a top toolbar with various simulation and modeling tools, a left sidebar with a library browser, and a central workspace displaying a Simulink block diagram. The diagram shows a sequence of blocks connected in a chain, starting with a 'Step' block, followed by several 'Simulink-PS Converter' blocks, and ending with an 'Actuator4' block. The status bar at the bottom indicates the simulation is running at 5:32 pm.

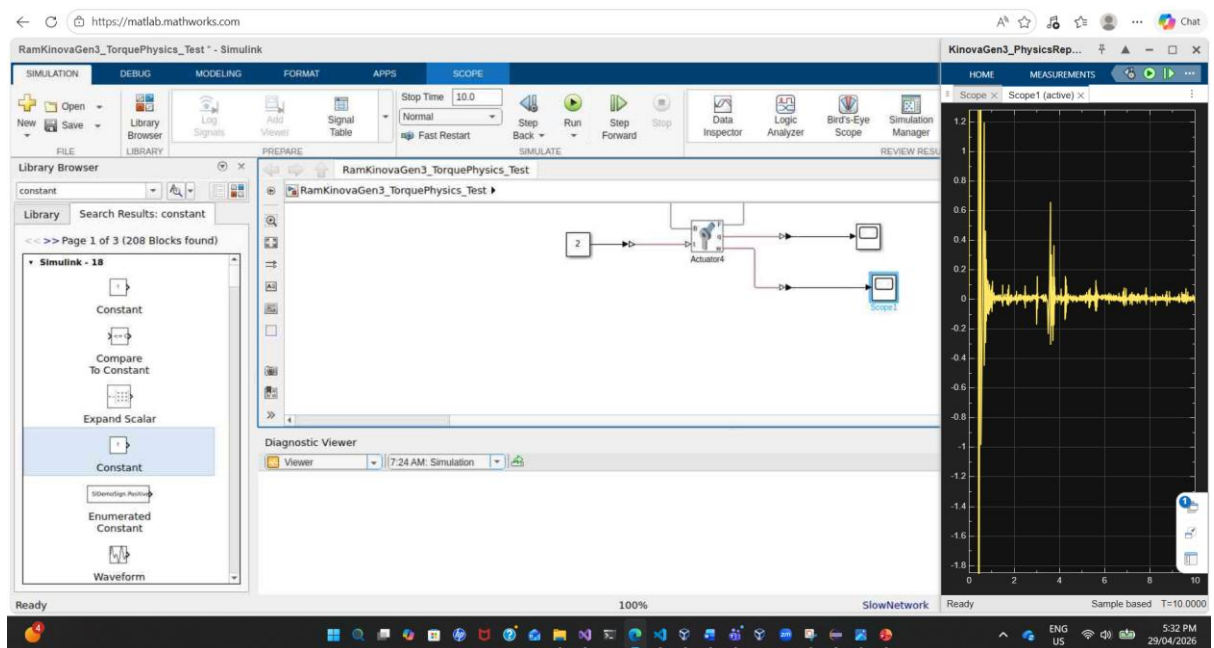
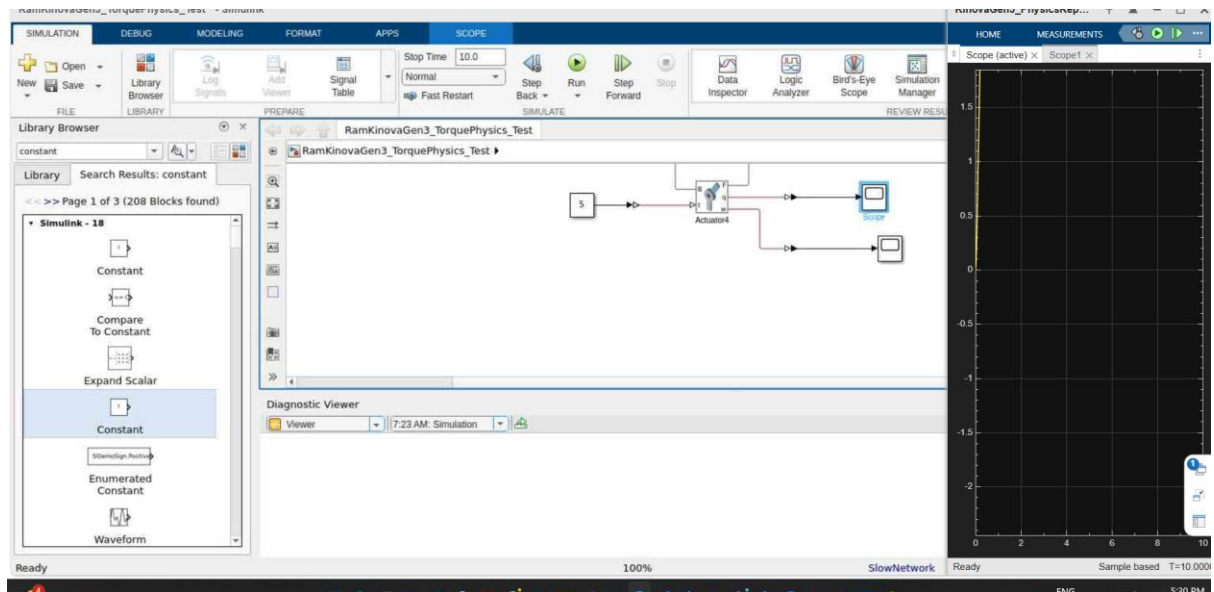


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VIDEO

<https://youtu.be/CutUs8sw0A0>

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